DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

FINAL YEAR MAJOR CAPSTONE PROJECT REPORT & DEFENSE HANDBOOK

ACADEMIC YEAR 2025 – 2026

MASTER EVALUATION DOSSIER & WORK DIVISION MATRIX

TERRITORY RUNNER

Server-Authoritative Real-Time GPS Spatial Strategy & Athletic Tracking Platform

THREE-STUDENT ENGINEERING PARTNERSHIP (33.3% / 33.3% / 33.3%)

MEMBER 1 (33.3%)
AYUSH
Backend, PostGIS & Anti-Cheat Lead

MEMBER 2 (33.3%)
GURUDUTT
Mobile Client & Offline Sync Lead

MEMBER 3 (33.3%)
AVINASH
Geometry, WebSockets & AI Lead

Candidate Representatives	Internal Project Guide	External Viva Examiner / HOD
AYUSH | GURUDUTT | AVINASH	Dept. of CSE	University Board

1. Master Subsystem Allocation & Responsibility Matrix

The **Territory Runner** platform is a multi-tier distributed spatial computing system. To ensure equitable rigor and clear accountability, the system was divided into three distinct, non-overlapping architectural domains of equal academic and engineering weight.

> **Work Equity Guarantee:** Each candidate owns approximately one-third of the codebase, possesses distinct theoretical foundations (Distributed Concurrency, Mobile Lifecycle Hardening, and Computational Geometry/AI), and is individually defensible under tough university examination standards.

Master Subsystem Distribution Matrix

Domain	AYUSH (Backend / DB)	GURUDUTT (Mobile / Offline)	AVINASH (Geometry / AI)
Primary Stack	Go 1.24, PostgreSQL 16, PostGIS 3.4, Docker	Flutter 3.13+, Dart 3, Hive Box Storage	Dart Clipper2, Go WebSockets, Gemini Flash, FCM
Core Modules Owned	• Chi REST Router & Middleware • PostGIS Spatial Schema & GiST • `pg_advisory_xact_lock` Engine • Kinematic Anti-Cheat Filter • Firebase RS256 JWT Verifier • Idempotency Deduplication	• Flutter Modular Architecture • Tactical Map Layer Rendering • Foreground Location Service • CPU WakeLocks (`wakelock_plus`) • Crash Recovery Safety Box • Persistent FIFO Sync Queue	• Lasso Loop Closure Snapping • Shoelace / Geodesic Parity • Ray-Casting Donut Classifier • WebSocket Radar Hub • Singleflight Coalescing • FCM HTTP v1 RS256 Push • Gemini 1.5 Flash AI Coach

Key Source Files	`backend/cmd/server/main.go`
	`backend/internal/api/`
	`backend/internal/store/postgres.go`
	`backend/internal/anticheat/`
	`backend/internal/auth/`
		`lib/main.dart`
		`lib/features/map/`
		`lib/services/location_service.dart`
		`lib/services/run_recovery_service.dart`
		`lib/services/sync_queue.dart`
			`lib/features/gameplay/polygon_*.dart`
			`backend/internal/geo/shoelace.go`
			`backend/internal/ws/hub.go`
			`backend/internal/notify/fcm.go`
			`backend/internal/coach/gemini.go`

Theoretical Viva Defense	ACID vs Spatial Consistency, Advisory Locks vs OCC/Redlock, OGC Simple Features, Biomechanical Velocity Limits	Android Doze & iOS Freezing, OOM Killer circumvention, Atomic Key-Value Disk I/O, UI Ergonomics	Computational Geometry, Planar vs Ellipsoidal Projections, Cache Stampedes, Singleflight, JWT Assertions
Test Contributions	79 Passing Go Unit/Integration Tests	317 Passing Flutter Unit/Widget Tests	15 Cross-Platform Golden Parity Tests
Contribution %	**33.3%**	**33.3%**	**33.3%**

RACI Responsibility Assignment Matrix

Legend: R = Responsible (executes), A = Accountable (final decision), C = Consulted (inputs), I = Informed (notified).

Lifecycle Feature / Architecture Step	AYUSH	GURUDUTT	AVINASH
High-Accuracy 1 Hz Location Capture & Battery Longevity	I	**R / A**	C
Real-Time Loop Closure Detection (Lasso Snapping)	C	C	**R / A**
Mobile Crash Recovery Safety Box (Hive Disk Serialization)	I	**R / A**	I
Optimistic Area Preview & Cross-Engine Golden Verification	C	C	**R / A**
Persistent FIFO Offline Sync Queue & Exponential Backoff	C	**R / A**	I
REST API Routing, Rate Limiting & Firebase RS256 Auth	**R / A**	I	C
Kinematic Anti-Cheat & GPS Spoofing Mitigation Filter	**R / A**	I	C
PostGIS Transaction Advisory Locking (`pg_advisory_xact_lock`)	**R / A**	I	I
PostGIS Spatial Topology (`ST_Union` , `ST_Difference`)	**R / A**	I	C
Real-Time WebSocket Radar & Singleflight Deduplication	C	C	**R / A**
Asynchronous FCM HTTP v1 Push Notification Alerts	C	I	**R / A**
Google Gemini 1.5 Flash AI Tactical Coaching Proxy	C	I	**R / A**
Weekly Competitive League Reset & Percentile Brackets	C	C	**R / A**

2. Consolidated Architecture & Cross-Member Data Pipeline

The sequence below illustrates how the three engineering domains integrate seamlessly during a real-world athletic run session:



3. Joint 20-Minute Defense Presentation Strategy

For the college capstone evaluation panel, the team should execute a synchronized 20-minute presentation followed by a 10-minute viva voce:

Time Window	Presenter	Presentation Content & Slide Focus	Visual Asset / Demo
0:00 – 2:00 (2 mins)	**AYUSH** (Team Lead)	• Introduction of project & teammates. • The problem with passive linear trackers (Strava). • Executive overview of Territory Runner & architecture.	Project banner & comparative analysis matrix.
2:00 – 7:30 (5.5 mins)	**AYUSH**	• Server-authoritative architecture & Go 1.24 Chi API. • PostGIS 3.4 spatial topology (`ST_Union` , `ST_Difference`). • Advisory Lock (`pg_advisory_xact_lock`) vs OCC/Redis. • Kinematic anti-cheat filter & spoofing mitigation. • Idempotency keys & Docker deployment.	Database schema, lock contention graph, anti-cheat math.
7:30 – 13:00 (5.5 mins)	**GURUDUTT**	• Flutter client architecture & Athletic Minimal design. • Foreground location service & CPU wakelocks. • Overcoming Android Doze & iOS background throttling. • Hive Crash Recovery Safety Box (zero lost waypoints). • Offline-first FIFO sync queue & 1400ms hold gesture.	Physical device demo, crash simulation & recovery.
13:00 – 18:30 (5.5 mins)	**AVINASH**	• Lasso loop closure snapping & segment intersection math. • Algorithmic parity: Shoelace vs PostGIS (< 0.5% variance). • Ray-casting donut hole classifier (Jordan Curve). • WebSocket radar & singleflight cache coalescing. • FCM HTTP v1 push & Gemini 1.5 Flash AI coaching.	Golden fixture comparison table & live radar view.
18:30 – 20:00 (1.5 mins)	**ALL THREE**	• Live integrated demonstration: simulating a run loop. • Test metrics: 317 Flutter tests + 79 Go tests passing. • Summary of results and transition to Viva examination.	Live dashboard & test execution badge.

4. College Examination Evaluation Rubric & Sign-Off

Suggested grading evaluation rubric for internal and external project assessment panels:

Evaluation Parameter	Max Marks	AYUSH (Backend)	GURUDUTT (Mobile)	AVINASH (Geometry/AI)
Architecture & Design Rigor	20	PostGIS schema, advisory locks, clean Chi middleware pipeline.	Flutter modular design, foreground service, Hive persistence.	Dual-engine parity, singleflight WebSocket hub, FCM RS256 flow.
Code Quality & Implementation	20	Go 1.24, strict concurrency control, zero race conditions.	Dart 3, clean reactive state, zero analyzer warnings.	Clipper2 integration, mathematical formulas, prompt engineering.
Testing, Parity & Verification	20	79 Go automated tests, lock stress test, CI pipeline.	317 Flutter tests, crash recovery simulation, battery profile.	15 cross-platform golden fixtures, WebSocket concurrency load tests.
Viva Voce & Technical Defense	20	Defense of distributed locks, ACID, anti-cheat kinematics.	Defense of OS lifecycle, Doze mode, offline sync mechanics.	Defense of Shoelace geometry, singleflight, FCM OAuth2.
Individual Contribution & Documentation	20	33.3% equal partner, individual defense dossier, clean docs.	33.3% equal partner, individual defense dossier, clean docs.	33.3% equal partner, individual defense dossier, clean docs.
TOTAL SCORE	**100**	**____ / 100**	**____ / 100**	**____ / 100**

CONSOLIDATED FINAL PROJECT EXAMINATION SIGN-OFF

The undersigned examination committee has evaluated the project work entitled **"Territory Runner — Server-Authoritative Real-Time GPS Spatial Strategy & Athletic Tracking Platform"** presented jointly and individually by **AYUSH**, **GURUDUTT**, and **AVINASH**. The individual contributions, technical defenses, and software deliverables meet the standards for the award of the Degree of Bachelor of Technology in Computer Science & Engineering.

Internal Project Guide Signature & Date	**External Viva Examiner** Signature & Date	**Head of Department (CSE)** Signature & Seal